

Sogand Tatlari

AI Researcher — Numerical Optimization, LLM Reliability & Systematic Failure Characterization

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RESEARCH FOCUS

Software engineer (B.Sc. Computer Engineering) working on AI reliability research — specifically, why LLMs fail in unpredictable ways depending on how they're prompted, and how better-designed optimizers can improve neural network training on tabular data. Looking for an MSc thesis position to pursue this research more formally.

EDUCATION

Hakim Sabzevari University — B.Sc. Computer Engineering

2017 – 2022

Full Scholarship, public government-funded university | GPA: 16.56/20.0 ($\approx 3.4/4.0$, B+/A-)

- Capstone (20/20): Domain-Specific Semantic Search Engine — NLP-based retrieval system outperforming keyword baselines on a specialized technical corpus.

RESEARCH EXPERIENCE

Independent AI Researcher

2024 – Present

Two co-authored preprints (2025), spanning optimizer design and LLM reliability research

LPECO: Control-Theory-Inspired Optimizer for Neural Networks on Tabular Data

- Designed LPECO (Lion with Predictive Error Correction), adding a PI-control-inspired integral error-correction term to the Lion optimizer to dampen excess momentum and reduce overshoot on non-convex tabular loss landscapes.
- Achieved formal statistical equivalence (TOST) to LightGBM and 98.4% Bayesian (ROPE) practical equivalence to XGBoost across an 8-dataset benchmark, with the highest mean rank among all methods (Friedman test).
- Ran a symmetric, TPE-guided two-stage hyperparameter search (128 trials/pairing) to ensure fair, reproducible comparison across optimizers and GBDT baselines.
- Identified a class-imbalance-sensitive performance pattern (Spearman's $\rho \approx 0.54$, not statistically significant at $N=8$) via post-hoc meta-analysis across datasets, generating a testable hypothesis for future work on imbalanced/low-resource tabular learning.
- Delivered a statistically dominant result (Mann-Whitney U, $p < 0.001$) on a safety-critical diagnostic dataset (BreastCancer), evidencing the real-world stakes of optimizer reliability.

Emergent Brittleness: Pathological Instruction-Following in LLM-Based Code Optimization

Under the supervision of Prof. H.R. Hamidi, Imam Khomeini International University (IKIU) — guest student

- Designed and ran a controlled $3 \times 3 \times 5$ factorial experiment (270 runs) across 3 LLMs, 3 prompt strategies, and 5 novel synthetic code-optimization tasks to isolate prompt–architecture interaction effects.
- Applied Generalized Linear Mixed-Effects Models (GLMM, binomial) for correctness and Generalized Least Squares (GLS) for performance, statistically confirming that elaborate 'expert' prompts significantly degrade correctness for a lightweight model ($p=0.030$), with a trend toward severe performance degradation for another model ($p=0.059$).
- Coined 'emergent brittleness' — a failure mode in which highly instruction-tuned LLMs show correlated correctness and performance degradation under specific, benign-looking prompt conditions — and proposed a Cognitive Load Theory account of the mechanism.
- Connected findings to broader AI-safety concerns (emergent misalignment, model collapse) and outlined a diagnostic/prescriptive research agenda for model-specific, empirically validated prompt engineering and human oversight of AI-generated code.
- Code & replication package: github.com/isogand/emergent-brittleness-llm-study

PUBLICATIONS

- S. Mobarhan*, S. Tatlari*. "Unlocking Potential: A Case for Neural Network Perseverance on Critical Tabular Tasks Through Principled Optimizer Design." Preprint, 2025. doi.org/10.5281/zenodo.17116288
- S. Tatlari*, S. Mobarhan*, H. R. Hamidi. "Emergent Brittleness: A Case Study on Pathological Instruction-Following in LLM-Based Code Optimization." Preprint, 2025. doi.org/10.5281/zenodo.17116501

*Denotes equal contribution.

PROFESSIONAL EXPERIENCE

Daric Data Processing (Tehran, Iran) — Mobile Developer & Front-End Team Lead

Apr 2023 – Present

- Led a team of front-end developers across project timelines and code quality for high-traffic web and mobile applications.
- Developed the company's primary cross-platform mobile app (React Native) for flight and hotel booking.
- Pioneered AI adoption within the company, independently designing and deploying production ML models — this hands-on exposure to model reliability and optimization challenges in production directly motivated my transition to independent AI research:
- Gender Recognition: computer vision model automating gender verification from national ID photos, improving processing efficiency by ~40%.
- Comment Moderation: NLP system detecting and flagging offensive language for a football club's website.

dreamworldestate — Mobile Developer

Dec 2021 – Mar 2023 | Türkiye (Remote-part time)

- Developed and maintained the company's React Native mobile application for real estate listings.

TECHNICAL SKILLS

- Data Science & ML: Python, PyTorch, Keras, TensorFlow, scikit-learn, pandas, NumPy, MLOps
- Statistical Methods: GLMM, GLS, factorial experimental design, TOST/Bayesian equivalence testing, Friedman/Mann-Whitney U tests, hyperparameter optimization (TPE)
- Domains: Natural Language Processing, Computer Vision, Generative AI (Diffusion Models), LLM Evaluation
- Web & Mobile: React Native, React, JavaScript, REST APIs

CERTIFICATIONS & VOLUNTEER EXPERIENCE

- Certifications: Natural Language Processing; [Custom and Distributed Training with TensorFlow](#) — DeepLearning.AI & Coursera